An improved upper bound on the Grothendieck constant of order three from symmetric local models

Claude Opus 5.5 Extra Effort[1]

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The Grothendieck constant of order three, $K_G(3)$, is the inverse of the critical visibility v_c below which the two-qubit Werner state admits a local hidden-variable model for all projective measurements. We prove $K_G(3) \le 1.4494177506$, improving the bound 1.4546336567 of Designolle *et al.* (2023); equivalently, the Werner state is local for projective measurements up to visibility 0.6899322156. With the known lower bound $K_G(3) \ge 1.43670$, the window for $K_G(3)$ shrinks by 29%. The proof is an explicit local model for 721 measurement directions, extended to all directions by a shrinking argument. It is verified in exact rational arithmetic and re-checked by an independent program. Three ingredients make it possible. Symmetrization under the icosahedral rotations and the exchange of the parties reduces the underlying linear program more than 110-fold. A pipeline of Frank–Wolfe pools, interior-point linear programming and primal-dual column generation solves the finite problems to within about 10^{-4} of heuristic estimates of their critical visibility. Geodesic direction sets nearly saturate the covering bound for the shrinking factor, and small non-unit deformations of the vectors, which the shrinking argument allows, add a little more. Along the way we certify critical visibilities of finite direction sets (≥ 0.69261 for 406 directions, ≥ 0.69230 for 721). Their slow decrease suggests $v_c \approx 0.692$, i.e. $K_G(3) \approx 1.445$. The local models were found by a multi-agent language-model system working against a frozen exact verifier, and all certificates are released with this work.

I. INTRODUCTION

Grothendieck's inequality [1] states that for every real matrix M and all unit vectors a_i, b_j in a real Hilbert space,

$$\sum_{ij} M_{ij}\, a_i \cdot b_j \;\le\; K_G \max_{x_i, y_j = \pm 1} \sum_{ij} M_{ij}\, x_i y_j, \qquad (1)$$

with a universal constant K_G. Restricting the vectors to $\mathbb{R}^d$ defines constants $K_G(d)$, which are non-decreasing in d. Only $K_G(2) = \sqrt{2}$ is known exactly [2]. In quantum information, $K_G(3)$ has a direct physical meaning: Acín, Gisin and Toner showed that the two-qubit Werner state

$$\rho_v = v|\psi^-\rangle\langle\psi^-| + (1-v)\,\mathbb{1}/4 \qquad (2)$$

admits a local hidden-variable model for all projective measurements if and only if $v \le v_c = 1/K_G(3)$ [3]. Werner's original model works up to $v = 1/2$ [4], and v_c remains unknown.

Bounds on v_c come from two kinds of explicit objects (Table I). Bell inequalities with exactly computed local bounds give upper bounds on v_c, i.e. lower bounds on $K_G(3)$ [5–10]. The best one is a 97×97 matrix from Ref. [10], whose local bound was computed exactly with a quadratic unconstrained binary optimization (QUBO) solver. With improved quantum vectors it gives $K_G(3) \ge 1.4367003433$ [11]. Local models for finitely many measurements, extended to all measurements by a shrinking argument, give lower bounds on v_c [10, 12–14]. Designolle *et al.* built such a model for 406 directions with Frank–Wolfe algorithms and obtained $v_c \ge 0.6874583132$, i.e. $K_G(3) \le 1.4546336567$ [10].

Here we prove

$$K_G(3) \le 1.4494177506, \qquad v_c \ge 0.6899322156. \qquad (3)$$

This lowers the upper bound by 5.22×10^{-3}. Together with the lower bound of Ref. [11], the window $[1.4367003433, 1.4494177506]$ is 29% narrower than before. The certificate is a local model on 721 directions. Its correctness reduces to exact rational arithmetic and an exact three-dimensional convex hull (Theorem 1). Our verifier checks it in seconds, and an independently written program re-checks it. Three ingredients get us there. (i) The direction sets have icosahedral symmetry. Averaging the local model over the 60 rotations and the exchange of the two parties reduces the linear program from m^2 to about $m^2/115$ coordinates for the sizes used here (Sec. III). (ii) Frank–Wolfe pools, interior-point linear programming and primal-dual column generation solve these programs to within about 10^{-4} of heuristic (non-rigorous) estimates of the critical visibility of the finite set (Sec. IV). (iii) Geodesic icosahedral spheres nearly saturate the covering bound for the shrinking factor. A small, dual-guided non-unit deformation of the vectors gives a further, smaller gain (Sec. V). We also report certified lower bounds on the critical visibilities of finite direction sets (Table II, Fig. 1). Their slow decrease with the number of directions suggests $v_c \approx 0.692$ (Sec. VII).

The search was carried out by a multi-agent system of language models working against a verifier that was written, tested and frozen before any search began (Sec. VIII).

TABLE I. Bounds on the critical visibility $v_c = 1/K_G(3)$ of the two-qubit Werner state under projective measurements (after Table I of Ref. [10]). "Inputs" is the number of measurement settings per party; $m \sim \infty$ means a local model on m directions extended to all directions by shrinking. Upper bounds on v_c are lower bounds on $K_G(3) = 1/v_c$ and vice versa. Where the exact values are available (CHSH and Refs. [10, 11, 14]), upper bounds are rounded up and lower bounds down; the other entries are as printed in Ref. [10]. Krivine's bound on $K_G(3)$ was translated into a Werner visibility in Ref. [3].

	v_c	reference	inputs
upper	0.70711	CHSH 1969 [5]	2
bounds	0.7056	Vértesi 2008 [6]	465
	0.7054	Hua *et al.* 2015 [7]	∞
	0.7012	Brierley *et al.* 2016 [8]	42
	0.6964	Diviánszky *et al.* 2017 [9]	90
	0.6961	Designolle *et al.* 2023 [10]	97
	0.69604	Designolle *et al.* 2026 [11]	97
lower	0.5	Werner 1989 [4]	∞
bounds	0.6595	Krivine 1979 [2, 3]	∞
	0.68289	Hirsch *et al.* 2017 [14]	$625 \sim \infty$
	0.68745	Designolle *et al.* 2023 [10]	$406 \sim \infty$
	0.68993	this work	$721 \sim \infty$

II. THE CERTIFICATE

For $m, n \geq 1$ let $\mathcal{L}_{m,n} = \mathrm{conv}\{xy^T : x \in \{\pm 1\}^m,\ y \in \{\pm 1\}^n\}$ be the local polytope of full correlators. A correlation matrix $T \in \mathcal{L}_{m,n}$ is reproduced by a local model with deterministic ± 1 outputs; pairing every deterministic strategy (x, y) with $(-x, -y)$ makes all marginals vanish. For projective measurements along unit vectors $a, b \in S^2$, the state ρ_v has $\langle A \rangle = \langle B \rangle = 0$ and $\langle AB \rangle = -v\, a \cdot b$. The sign can be absorbed by Bob relabelling his outcomes, so ρ_v is local for all projective measurements if and only if $v\,(a_i \cdot b_j)_{ij} \in \mathcal{L}_{n,n}$ for all n and all unit vectors. The "if" direction needs a compactness argument; together with Eq. (1) for $d = 3$ this gives $v_c = 1/K_G(3)$ [3]. Theorem 1 below builds the model for all measurements directly and does not use this equivalence.

The following lemma is Lemma 1 of Ref. [10], stated there for $m = n$. We include its short proof.

Lemma 1. *The Frobenius unit ball* $\{E \in \mathbb{R}^{m \times n} : \|E\|_F \leq 1\}$ *is contained in* $\mathcal{L}_{m,n}$.

Proof. Let x, y be independent and uniformly random sign vectors. For every Y, $\mathbb{E}[(x^T Y y)^2] = \sum_{ij} Y_{ij}^2 = \|Y\|_F^2$, so $\max_{x,y} x^T Y y \geq \|Y\|_F$ (replacing x by $-x$ flips the sign). The support function of $\mathcal{L}_{m,n}$ therefore dominates that of the ball, and both sets are closed and convex. □

Let $\eta(X)$ denote the inradius of $\mathrm{conv}\{\pm x_1, \ldots, \pm x_m\}$ for vectors $x_i \in \mathbb{R}^3$, i.e. the largest η such that this polytope contains the ball of radius η about the origin. It is positive when the x_i span $\mathbb{R}^3$. We write U and V for the matrices with rows u_i and w_j.

Theorem 1. *Let* $u_1, \ldots, u_m$ *and* $w_1, \ldots, w_n$ *be vectors spanning* $\mathbb{R}^3$, *let* $\lambda > 0$ *and* $T_{ij} = \lambda\, u_i \cdot w_j$. *Suppose there are weights* $p_k \geq 0$ *and sign vectors* $s_k \in \{\pm 1\}^m$, $t_k \in \{\pm 1\}^n$ *such that*

$$\sum_k p_k + \Big\| T - \sum_k p_k s_k t_k^T \Big\|_F \leq 1. \tag{4}$$

Then ρ_v *is local for all projective measurements whenever* $v \leq \lambda\, \eta(U)\, \eta(V)$, *and* $K_G(3) \leq 1/(\lambda\, \eta(U)\, \eta(V))$.

Proof. Let $\omega = \sum_k p_k$ and $E = T - \sum_k p_k s_k t_k^T$. Then $T = \sum_k p_k s_k t_k^T + \|E\|_F\, (E/\|E\|_F) + (1 - \omega - \|E\|_F) \cdot 0$ (the middle term is absent if $E = 0$) is a convex combination of points of $\mathcal{L}_{m,n}$, by Lemma 1 and because $0 \in \mathcal{L}_{m,n}$. Hence $T = \sum_\ell q_\ell\, x^{(\ell)} y^{(\ell)T}$ with a probability vector q. Let $a \in S^2$. Since $\eta(U)\, a \in \mathrm{conv}\{\pm u_i\}$, there are $\alpha_i^\pm \geq 0$ with $\sum_i (\alpha_i^+ + \alpha_i^-) = 1$ and $\eta(U)\, a = \sum_i (\alpha_i^+ - \alpha_i^-) u_i$, and likewise $\beta_j^\pm$ for $\eta(V)\, b$. In the local model, the shared variable ℓ is drawn with probability q_ℓ. Alice draws $(i, \pm)$ with probability $\alpha_i^\pm$ and outputs $\pm x_i^{(\ell)}$; Bob does the same with β and $y^{(\ell)}$. Then

$$\mathbb{E}[A_a B_b] = \sum_{ij} (\alpha_i^+ - \alpha_i^-)(\beta_j^+ - \beta_j^-)\, T_{ij} = \lambda\, \eta(U)\, \eta(V)\ a \cdot b,$$

and a global sign flip with probability $1/2$ removes the marginals. Mixing with white noise lowers the visibility to any $v \leq \lambda \eta(U) \eta(V)$. For the bound on $K_G(3)$, evaluate any Bell functional on this model. For all M and unit $a_i, b_j \in S^2$, $\lambda \eta(U) \eta(V) \sum_{ij} M_{ij} a_i \cdot b_j = \mathbb{E} \sum_{ij} M_{ij} A_{a_i} B_{b_j} \leq \max_{x,y} \sum_{ij} M_{ij} x_i y_j$, because the expectation is an average of deterministic values. □

Three remarks on Theorem 1. (a) Ref. [10] uses the special case $u = w$ with unit vectors and $\lambda = \nu_2 v_0$, where $\nu_2 = 1/(1 + \|x_T - v_0 U U^T\|_F)$ and x_T is the Frank–Wolfe iterate. In the notation of the theorem, $\sum_k p_k = \nu_2$ and $\|E\|_F = 1 - \nu_2$. Here the two parties may use different sets, and the vectors need not be unit. Only the inradius of $\mathrm{conv}\{\pm u_i\}$ and the locality of $\lambda U V^T$ matter, and rescaling all u_i by c and λ by $1/c$ changes nothing. We exploit this freedom in Sec. V. (b) Replacing a term $s_k t_k^T$ by its average over any list of signed index permutations, $(h \cdot X)_{ij} = \sigma_h^A(i) \sigma_h^B(j) X_{\pi_h^A(i) \pi_h^B(j)}$, or by its transpose (for $m = n$), keeps it in $\mathcal{L}_{m,n}$, because every image of a deterministic strategy is deterministic. The list need not be a group, and T need not be invariant. This is how symmetrized models are certified. (c) A local model for all projective measurements at visibility v yields a local model for all positive-operator-valued measurements (POVMs) at visibility $2v/3$ [10, 14, 15]. Hence Eq. (3) gives $v_c^{\mathrm{POVM}} \geq 0.45995$, compared with 0.45830 from Ref. [10].

Exact verification. A certificate (format in Appendix A) lists rational vectors u_i, w_j, a rational λ, positive integer weights n_k over a common denominator D, the sign vectors as bit strings, and optionally a list of signed permutations and a transposition flag. The verifier accumulates $N = \sum_k n_k s_k t_k^T$ and its symmetrization in exact 64-bit integer arithmetic, with caps that rule out overflow. It bounds every entry of E from above by a multiple of 2^{-64}, computes both inradii from an exact rational convex hull that it validates independently, and compares $1/(\lambda\eta(U)\eta(V))$ with the record exactly, on squares of rationals. Floating-point arithmetic is used only to skip exact checks for points far inside a facet plane (Appendix A); no verdict depends on it.

III. SYMMETRY REDUCTION

All direction sets of Table II are invariant, as sets of antipodal pairs, under the rotation group I of the icosahedron (order 60). (The 97-direction set used for a timing comparison in Sec. IV has octahedral symmetry instead.) Each rotation g acts on the pairs by a signed permutation, $gu_i = \sigma_g(i)\, u_{\pi_g(i)}$. For unit vectors, and for vectors whose length depends only on the orbit, the target $T = \lambda UU^T$ is invariant under the induced action $(g \cdot T)_{ij} = \sigma_g(i)\sigma_g(j)T_{\pi_g(i)\pi_g(j)}$ and under transposition. If $T \in \mathcal{L}$, averaging any decomposition over the group of order 120 generated by I and transposition gives a decomposition into *orbit atoms* $\mathrm{Sym}(st^T)$, i.e. averages of the 120 images of one deterministic strategy. The linear program therefore lives on the invariant subspace, whose coordinates are the orbits of index pairs. Their number R (Table II) approaches $m^2/120$ for large m. For the sizes used here, m^2/R ranges from 112 to 115: $R = 1469$ for $m = 406$ instead of $406^2 = 164{,}836$, and $R = 4507$ for $m = 721$. Pricing a new atom against a dual vector y on the orbit coordinates amounts to maximizing the bilinear form $s^T\bar{Y}t$ over sign vectors, where $\bar{Y}$ is the full matrix obtained by spreading y over the orbits.

Symmetrized full-correlation polytopes were studied in Ref. [16] for symmetries that permute the parties. Frank–Wolfe methods on symmetrized correlation polytopes, for signed permutation groups acting on the settings, were introduced in Ref. [17]. The polyhedra of Ref. [10] were already icosahedral, but its record local model was computed without symmetrization, with 78,747 deterministic strategies. Each of our orbit atoms stands for up to 120 deterministic strategies. The final certificate has 19318 orbit atoms, i.e. up to 2,318,160 deterministic strategies, on a linear program with 4507 coordinates.

Rational vectors cannot have exact icosahedral symmetry, since a rotation of order five has the irrational trace $(1+\sqrt{5})/2$. We compute with floating-point symmetric sets and certify with exact rational vectors close to them (the final certificate's vectors are symmetric up to 5×10^{-15}). The index permutations stay exact. Theorem 1 does not require symmetry, and the tiny asymmetry of the rational target ends up in the residual E.

IV. SOLVING THE FINITE PROBLEMS

For a set X of m unit vectors, let $v_c(X) = \max\{v : vXX^T \in \mathcal{L}_{m,m}\}$ be its critical visibility. Since any infinite model restricts to X, $v_c \le v_c(X)$. Theorem 1 gives $v_c \ge \lambda\eta(X)^2$ with $\lambda \le v_c(X)$. The task is therefore to certify λ as close to $v_c(X)$ as possible, for sets X that are as large as is computationally affordable.

Linear minimization oracle. Every method below needs atoms that maximize a linear functional, $\max_{s,t} s^T\bar{Y}t$. This is a bipartite max-cut (QUBO) problem and NP-hard. We solve it heuristically: alternating exact updates $t \leftarrow \mathrm{sign}(\bar{Y}^T s)$, $s \leftarrow \mathrm{sign}(\bar{Y}t)$ from many starts (random, spectral, hemisphere strategies $s_i = \mathrm{sign}(x_i \cdot r)$ for random r, and warm starts from the current atoms), followed by single-flip local search and, in the final stages, iterated local search [18]. A weak oracle only slows convergence. Every atom is a valid strategy, and the certificate does not depend on the oracle.

Frank–Wolfe pools at fixed v_0. Following Ref. [10], we first minimize $\frac{1}{2}\|x - v_0XX^T\|_F^2$ over the symmetrized polytope for a fixed v_0. The Frobenius norm is evaluated on the orbit coordinates with the orbit sizes as weights. We use a fully corrective variant [19–21]: after each oracle round the weights on all active atoms are re-optimized. We found that this restricted master problem, not the oracle, limits progress. In the first run a spectral projected-gradient method (Barzilai–Borwein steps [22] with exact line search on the simplex) built the pools; later pools were grown with an affine-scaling Levenberg–Marquardt master inside the Frank–Wolfe loop. That master solves $x = v_0XX^T$ on the pool to a residual of 10^{-9}–10^{-14} whenever the pool supports v_0. Even with a simpler accelerated projected-gradient master, the fully corrective method reached $v_0 = 0.69$ at $m = 97$ with residual 7×10^{-5} in about a minute, whereas plain simplex column generation reached only 0.615 in four minutes. On the 406-direction set of Ref. [10] it solves $v_0 = 0.692$, the value used there, to a residual of 2×10^{-10}, so that $\nu \approx 1 - 10^{-9}$ instead of $\nu_2 = 0.999914$ (from the supplementary file of Ref. [10]).

Linear programming over a pool. Given a pool of atoms, $\max\{v : Ap = vc,\ \mathbf{1}^T p = 1,\ p \ge 0\}$ (A: atoms in orbit coordinates, c: the target) is a dense linear program with $R+1$ rows. We solve it with a Mehrotra predictor-corrector interior-point method [23] with an ℓ_1-penalized slack. For a pool that solves v_0 exactly, the result satisfies $v_{\mathrm{pool}} \ge v_0$, and in practice it is larger. For example, it raised the 406-direction pool built at $v_0 = 0.692$ to $v_{\mathrm{pool}} = 0.69236$ (certified $\lambda \ge 0.69234$ after rounding). The dominant cost is one dense rank-K update of the $(R+1) \times (R+1)$ normal matrix per iteration.

Primal-dual column generation. The interior-point master supplies duals from the interior of the optimal face rather than a vertex, as in primal-dual column gen-

eration [24] (we used a fixed tolerance). New atoms are priced at these duals, with dual smoothing [25], and added in batches of up to a few thousand per round. The pool is trimmed to a cap of 12–22 thousand atoms by smallest slack. For the largest sets, a box-step stabilization [26] confines the duals to a box around a stable centre, and the true value is re-measured on the unboxed master every few rounds. Pools built by different methods were merged and re-solved, which repeatedly gave small gains. We stopped when a round improved v by about 10^{-6} or less.

From a pool to a certificate. The weights are rounded down to multiples of 2^{-46}. With ω their sum and δ the Frobenius norm of the residual, the common denominator is enlarged to $D = \lceil 2^{46}(\omega+\delta+10^{-9})\rceil$, which implements the analyticity factor $\nu \approx 1/(1+\delta)$ of Ref. [10]. Theorem 1's condition then holds with a margin of about 10^{-9}. The interior-point iterate leaves a residual $\|Ap - vc\|$ of about 10^{-5}, which costs about as much in λ. From board post 104 on (24 of the 41 certificates, including all of the best ones in Table II except pol406, pol406$_{1/4}$ and pol541), a few Newton steps first project p onto $\{Ap = vc,\ \mathbf{1}^T p = 1\}$ in the metric $\mathrm{diag}(p)^{-1}$, keeping all weights positive and lowering the residual to 10^{-10}–10^{-9}.

Heuristic dual estimates. With the oracle's best value h, $h/\langle \bar{Y}, c\rangle$ estimates $v_c(X)$ from above. It is not a bound, because the oracle may miss the true maximum. On converged pools these estimates exceeded the certified λ by 2×10^{-5}–1.4×10^{-4}. A stronger oracle at the same duals raised them further, at $m = 606$ to 2.4×10^{-4} above λ, so the gap $v_c(X)-\lambda$ is not controlled rigorously.

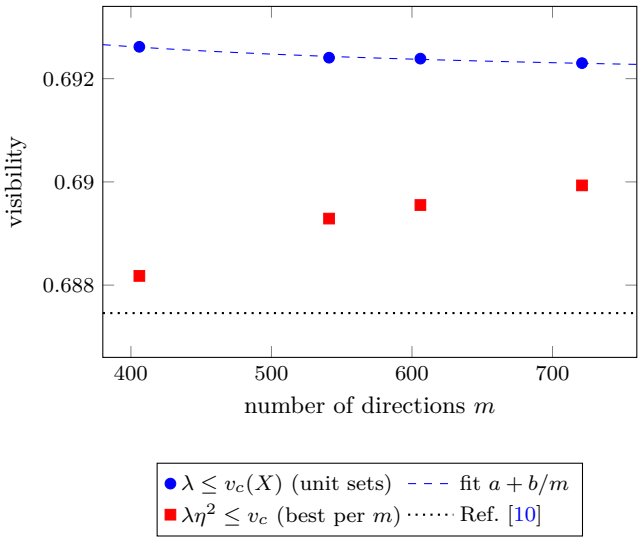


FIG. 1. Certified visibilities. Circles: certified lower bounds λ on the critical visibility $v_c(X)$ of the unit-vector sets pol406, pol541, geo606 and geo721 of Table II, with a least-squares fit $\lambda = a + b/m$ (dashed; $a = 0.6919$). Squares: best certified lower bound $\lambda\eta^2$ on v_c for each m, including the non-unit sets. The dotted line is the previous record $v_c \ge 0.6874583132$ [10]. The gap between circles and squares is approximately $\lambda(1-\eta^2)$, the cost of shrinking.

V. DIRECTION SETS

The shrinking factor $\eta(X)^2$ enters the bound linearly, so the direction set should cover the sphere as evenly as possible. The 406-direction polyhedron of Ref. [10] and the 541-direction polyhedron distributed with the BellPolytopes.jl package [27] are geodesic icosahedral spheres. Their 812 and 1082 vertices (406 and 541 antipodal pairs) are combinatorially those of GP$(9,0)$ (class I, $m = 5a^2+1$) and GP$(6,6)$ (class II, $m = 15a^2+1$) [28]. Ref. [10] built the former from two successive frequency-3 subdivisions. Starting from the projected subdivision, we optimized the free orbit parameters by sequential linear programming on the worst facet distances. This gives sets with $1-\eta^2 \approx 2.48/m$, close to the value $3^{-3/2}\cdot 4\pi/m \approx 2.42/m$ obtained by covering the sphere with $2m$ equal hexagonal cells, the asymptotically thinnest covering of the plane [29]. In particular, no rearrangement at fixed m can gain much. Random searches over orbit compositions did markedly worse than polished geodesic starts. We used the polished GP$(11,0)$ ($m = 606$) and GP$(12,0)$ ($m = 721$) spheres (geo606, geo721). We rationalize the vectors with an inverse stereographic projection, which, like the parametrization of Ref. [10], yields rational points exactly on the unit sphere.

Non-unit vectors. Theorem 1 allows each orbit o to have its own length $1+\epsilon_o$ and a slightly moved representative. Lengthening a vector enlarges the polytope near it and so can raise η, but it also makes the target harder to reproduce locally, so it lowers λ. Starting from a pool on the undeformed set, we took sequential-linear-programming steps in the orbit lengths and positions. The steps were guided by the interior-point duals, whose orbit components give the first-order change of $\ln v_{\mathrm{pool}}$. Each step was scored by the quantity the certificate actually pays, $v_{\mathrm{pool}}\,\eta^2/(1+\delta)$, after an exact re-solve on the pool, and the pool was then re-converged on the deformed set. On pol541, which is not polished, three accepted steps in two rounds raised η^2 from 0.995337 to 0.996231 at a cost of 5×10^{-4} in λ, a net gain of 2.3×10^{-4} in $K_G(3)$. On the polished geo721 the effect was much smaller. A step with $|\epsilon_o| \le 8\times10^{-5}$ raised η^2 from 0.996560 to 0.996691977 but lowered λ almost as much, a net gain of about 2×10^{-6} in $K_G(3)$ at equal pool effort. The final certificate's remaining advantage over the best unit-vector certificate, 2.6×10^{-5}, comes mostly from further pool optimization on the deformed set (rad721). Polished geodesic sets thus appear to be nearly stationary under such deformations.

TABLE II. Best certified bound per direction set. m: number of directions (antipodal pairs); R: orbit coordinates after symmetry reduction (icosahedral rotations and transposition); η^2: squared inradius of $\mathrm{conv}\{\pm u_i\}$, computed exactly from the certificate's vectors; λ: certified scale, which for the unit-vector sets is a certified lower bound on the set's critical visibility; $v = \lambda\eta^2$: certified Werner visibility; $K_G(3) \le 1/v$. Lower bounds are rounded down and upper bounds up. $\mathrm{pol406}_{1/4}$ is pol406 moved a quarter of the way to its polished optimum (same orbits and index order); rad541 and rad541′ are pol541 after one and after three per-orbit radius/position steps; rad721 is geo721 after one such step (Sec. V). Last column: board post of the certificate (Appendix B).

set	m	R	η^2	λ	$v = \lambda\eta^2 \ge$	$K_G(3) \le$	post
pol406	406	1469	0.993522757	0.6926171	0.6881309293	1.4532118198	35
$\mathrm{pol406}_{1/4}$	406	1469	0.993612507	0.6926023	0.6881783740	1.4531116318	42
pol541	541	2590	0.995337675	0.6924063	0.6891780786	1.4510037842	95
rad541	541	2590	0.995920092	0.6920968	0.6892731700	1.4508036051	104
rad541′	541	2590	0.996231172	0.6918967	0.6892891242	1.4507700250	115
geo606	606	3202	0.995904003	0.6923879	0.6895519102	1.4502171412	140
geo721	721	4507	0.996560717	0.6923008	0.6899198716	1.4494436834	213
rad721	721	4507	0.996691977	0.6922221	0.6899322156	1.4494177506	241

VI. RESULTS

Table II lists the best certificate for each direction set, and Fig. 1 shows the trends. On the 406-direction set of Ref. [10] we certify $\lambda \ge 0.6926171$, against $v_0\nu_2 = 0.69194$ there. This alone gives $K_G(3) \le 1.45322$. The critical visibilities of the unit-vector sets decrease only slowly with m: we certify $v_c(X) \ge 0.6926171$, 0.6924063, 0.6923879 and 0.6923008 for $m = 406$, 541, 606 and 721. Heuristic dual estimates exceed these by up to about 2×10^{-4} (Sec. IV). The shrinking factor improves much faster with m, so larger sets win.

Theorem 2. *The rad721 certificate (board post 241, SHA-256 prefix* `9f35fedc`*, see Data availability; 721 directions, $\lambda \ge 0.69222210$, $\eta^2 \ge 0.996691977$) satisfies the hypotheses of Theorem 1. Hence $K_G(3) \le 1.4494177506$, and the two-qubit Werner state ρ_v admits a local hidden-variable model for all projective measurements for $v \le 0.6899322156$, and for all POVMs for $v \le 0.45995$.*

The certificate uses the rad721 set (721 vectors with lengths within 8×10^{-5} of 1), 19318 orbit atoms averaged over the 60 icosahedral rotations and transposition, $D \approx 2^{46}$, residual $\|E\|_F \le 2.31 \times 10^{-10}$, and a hull with 2880 facets. The verifier checks it in under five seconds. The independent program of Sec. VIII agrees on every quantity.

VII. DISCUSSION

Where is v_c? The certified finite-set visibilities fall by only 3×10^{-4} from $m = 406$ to $m = 721$. A fit $\lambda(m) = a + b/m$, motivated by the $1/m$ scaling of the covering radius squared, extrapolates to $a = 0.6919$, i.e. $K_G(3) \approx 1.4453$. This is not a bound. The $\lambda(m)$ are lower bounds on $v_c(X)$ whose gaps are not uniform, the functional form is a guess, and $v_c(X)$ need not converge to v_c in this way. Still, it suggests $v_c \approx 0.692$, compared with the non-rigorous numerical estimate $v_c \approx 0.691$ obtained from gradient-trained local models with continuous measurement settings [30]. Either way, the present upper bound on $K_G(3)$ is probably within about 5×10^{-3} of the true value, while the lower bound of Ref. [11] is further away.

The lower bound. The duals of our linear programs are explicit Bell inequalities on the finite sets. Any Bell inequality on the 406-direction set has a ratio of quantum to local value of at most $1/v_c(X) \le 1/\lambda \le 1.4438$. With the local bound estimated by our heuristic oracle, the dual of the converged 406-direction program has ratio $\approx 1/0.6927 \approx 1.4436$. This is an upper estimate, since the true local bound can only be larger. Certifying that local bound exactly would therefore raise the lower bound from 1.4367 to at most about 1.4436, possibly less. It is a 406-variable dense QUBO with icosahedral symmetry, beyond the instances current exact solvers handle; Ref. [11] notes that symmetric instances are especially hard for branch and bound. Exploiting the symmetry in an exact solver would be a natural next step.

Further gains on the upper bound. Since covering is nearly saturated, further progress needs larger m. With $\lambda \approx 0.692$, polished geodesic spheres with $m = 846$ and 981 ($R = 6\,162$ and $8\,256$; constructed but not run) would give about 1.4493 and 1.4487. The cost is dominated by the dense interior-point master, which grows as R^2K.

Robustness. The certificate depends on no solver accuracy. Theorem 1's hypotheses are finitely many exact statements, and the verifier checks them in exact arithmetic, validating the hull independently of how it was computed. Every accepted certificate was also re-checked by a second program written independently of the verifier. That program has its own integer accumulation, a Qhull-based hull with exact repair and exact facet validation, and a rigorously rounded floating-point bound on the residual.

VIII. HOW THE RESULTS WERE FOUND

The work used two runs of a multi-agent system of language-model agents (Claude Sonnet 5.5, Anthropic). The agents ran as non-interactive coding sessions in 40-minute shifts on one desktop computer with ten cores and 16 GB of memory. A supervising session (Claude Opus 5.5) first built the verifier, its tests, the reference values and a baseline pipeline. It then coordinated the agents and audited their results. The agents shared a message board. They were not permitted to edit the verifier, the data or the board software: editing tools were denied for those paths, and the files were read-only. The verifier was hash-pinned, and the board rejected results reported with a modified verifier. Because this protection is not airtight, every result was re-checked independently. All 41 submissions improved on the reference value in force during their run. The agents screened every candidate with the verifier before submitting.

First run (four agents, about 55 minutes). One agent ported the Frank–Wolfe approach of Ref. [10] to the symmetry-reduced setting. It identified the restricted master as the bottleneck and produced the first improvement 20 minutes after launch, on the 406-direction set at $v_0 = 0.692$. A second agent wrote the interior-point linear program. Polishing the first agent's pools with it, and then column generation from the polished pools, gave the next four improvements within 20 minutes, the fourth on a partly polished 406-direction set. A third agent identified the polyhedra as geodesic spheres and built the larger sets. The fourth agent, the auditor, searched the literature, wrote out the argument of Theorem 1, re-checked every certificate with an independent program, and red-teamed the verifier. It also found that the supervising session's description of the win threshold was a factor of ten too strict. The verifier's own exact test was correct, so no verdict was affected. The first run produced 6 improvements, the best $K_G(3) \le 1.45294$ on 541 directions. It ended after about 55 minutes, when the computer hung, most likely under memory pressure, and rebooted.

Second run (three agents, four hours). The reference value was raised to the first run's best, and a memory guard watched the agents' jobs. It stopped jobs three times: twice for exceeding the per-job memory cap, and once at moderate memory use (swap above a 6 GB threshold while 35% of memory was free), after which the threshold was raised and jobs were measured by their memory footprint. The second run produced 35 improvements. The first, on 541 directions, came one minute after launch, from a pool left by the first run; the first on 606 directions came after 9 minutes; the first below 1.45, on 721 directions, after 2 hours 11 minutes. Pool unions, iterated local search and continued column generation on the deformed 721-direction set gave the last gains, and the final certificate came after 3 hours 38 minutes. The supervising session re-checked every improvement with the auditor's independent program; all 41 checks agree with the verifier (Appendix B).

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DATA AVAILABILITY

The certificate of Theorem 2 (exact JSON, 8 MB, SHA-256 `9f35fedc7b94778c1b7b77bd592f316d cad4e969d09a8e9414c93ee50908da9d`), all other certificates of Appendix B, the verifier with its tests, the independent checker, the direction sets and the search programs are released with this work. Tables II and III are generated from the certificates by a script in the repository.

Appendix A: Certificate format and verification

A certificate is a JSON object with: rational 3-vectors u_i (and w_j, or the same set), a rational scale λ, a denominator $D \le 2^{50}$, a list of strategies (n_k, s_k, t_k) with integers $n_k \ge 1$, $\sum_k n_k \le D$ and sign vectors encoded as hexadecimal bit masks, an optional list of G signed permutations $(\pi_h^A, \sigma_h^A, \pi_h^B, \sigma_h^B)$, and an optional transposition flag. The model is $L = \frac{1}{G(1+\mathrm{tr})} \sum_h [h \cdot N + (h \cdot N)^T_{\text{if tr}}]/D$ with $N = \sum_k n_k s_k t_k^T$, and the verifier checks the following.

1. *Accumulation.* N and its symmetrization are computed in C with 64-bit integers. Because $|N_{ij}| \le \sum_k n_k \le 2^{50}$ and $G(1+\mathrm{tr}) \le 4096$, no partial sum exceeds 2^{62}. Random entries of both are recomputed in Python as a guard.

2. *Residual.* For every entry, $|E_{ij}| = |T_{ij} - L_{ij}|$ is computed as an exact rational and rounded *up* to a multiple of 2^{-64}. The test $\sum_k n_k/D + \|E\|_F \le 1$ is then decided exactly on integers.

3. *Hull.* The convex hull of $\{\pm u_i\}$ is computed by a randomized incremental algorithm with a conflict graph and exact rational orientation tests. It is then validated independently: every directed edge has exactly one reverse; every input point lies on the inner side of every facet plane (exactly, after a floating-point screen that only skips points more than 10^{-9} inside); generic rays from the origin meet

exactly one facet, so the facets cover the boundary once; the origin is strictly inside; and the Euler characteristic is 2. η^2 is the minimum over facets of the squared distance to the origin, an exact rational.

4. *Verdict.* $1/(\lambda^2\eta(U)^2\eta(V)^2)$ is compared exactly with the square of the record minus 10^{-6}. A certified bound below the proven lower bound 1.4367003433 would raise an alarm.

The tests compare the hull with brute-force facet enumeration on random point sets, including degenerate ones. They reproduce the published shrinking factors of the 16- and 46-direction (32- and 92-vertex) polyhedra of Ref. [10] to 10^{-11}, compare the accumulation with a pure-Python implementation, and exercise every verdict except the timeout. The verifier code was frozen before the first run and not changed afterwards. Between runs only the reference value was raised.

Appendix B: All certificates

TABLE III: All certificates accepted by the verifier, in order of submission. Each improved on the reference value in force during its run (first run: the published 1.4546336567; second run: the first run's best, 1.4529358988), not necessarily on the previous certificate. Post: board post number; time: 2026 date and time; agent: the submitting agent; K: number of orbit atoms (each stands for up to 120 deterministic strategies); λ rounded down; audit: the independent re-check agrees (`results/audits.json`).

post	time	agent	set	K	λ	$K_G(3) \le$	audit
19	10-14 01:11	fw	pol406	14590	0.6919999	1.4545079069	yes
23	10-14 01:13	cg	pol406	14590	0.6923498	1.4537728277	yes
28	10-14 01:15	cg	pol406	18035	0.6924251	1.4536147936	yes
35	10-14 01:22	cg	pol406	11244	0.6926171	1.4532118198	yes
42	10-14 01:28	cg	$\mathrm{pol406}_{1/4}$	11096	0.6926023	1.4531116318	yes
45	10-14 01:30	fw	pol541	13239	0.6914855	1.4529358988	yes
61	10-15 00:18	cg	pol541	14933	0.6915622	1.4527746963	yes
69	10-15 00:26	fw	geo606	13028	0.6912293	1.4526479453	yes
73	10-15 00:30	cg	pol541	10000	0.6918409	1.4521895022	yes
75	10-15 00:30	geometry	rad541	10000	0.6914947	1.4520667720	yes
79	10-15 00:39	cg	pol541	14853	0.6922333	1.4513663569	yes
81	10-15 00:45	geometry	rad541	17411	0.6918847	1.4512484318	yes
83	10-15 00:47	fw	geo606	20000	0.6916999	1.4516594543	yes
89	10-15 00:48	cg	pol541	9735	0.6923760	1.4510672753	yes
91	10-15 00:52	geometry	rad541	9587	0.6920564	1.4508884117	yes
95	10-15 00:53	cg	pol541	16104	0.6924063	1.4510037842	yes
99	10-15 00:58	geometry	rad541	9862	0.6920890	1.4508200637	yes
104	10-15 01:06	geometry	rad541	9862	0.6920968	1.4508036051	yes
115	10-15 01:18	geometry	rad541′	17059	0.6918967	1.4507700250	yes
131	10-15 01:41	cg	geo606	8170	0.6923676	1.4502596307	yes
140	10-15 01:53	geometry	geo606	12091	0.6923879	1.4502171412	yes
167	10-15 02:29	fw	geo721	13818	0.6921911	1.4496734555	yes
175	10-15 02:30	geometry	geo721	17185	0.6920168	1.4500386910	yes
179	10-15 02:32	geometry	geo721	13267	0.6922253	1.4496018282	yes
184	10-15 02:39	geometry	geo721	16047	0.6922540	1.4495417312	yes
186	10-15 02:42	geometry	geo721	17377	0.6922613	1.4495263805	yes
190	10-15 02:45	geometry	geo721	17800	0.6922623	1.4495244125	yes
195	10-15 02:53	cg	geo721	11319	0.6922757	1.4494963268	yes
201	10-15 02:59	geometry	rad721	11319	0.6921853	1.4494947630	yes
205	10-15 03:07	geometry	rad721	16000	0.6921984	1.4494672680	yes
207	10-15 03:11	fw	geo721	16000	0.6922963	1.4494531967	yes
209	10-15 03:13	cg	geo721	10891	0.6922901	1.4494660745	yes
211	10-15 03:19	geometry	rad721	16000	0.6922061	1.4494510964	yes
213	10-15 03:20	fw	geo721	16000	0.6923008	1.4494436834	yes
216	10-15 03:21	cg	geo721	15893	0.6922979	1.4494498252	yes
218	10-15 03:22	geometry	rad721	17422	0.6922107	1.4494415778	yes
222	10-15 03:30	geometry	rad721	18292	0.6922178	1.4494266858	yes
224	10-15 03:33	cg	rad721	17255	0.6922180	1.4494261826	yes
231	10-15 03:43	fw	rad721	16000	0.6922210	1.4494200059	yes
236	10-15 03:51	geometry	rad721	18592	0.6922218	1.4494183326	yes
241	10-15 03:55	cg	rad721	19318	0.6922221	1.4494177506	yes

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